

# PRITHVIRAJ MODY

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## EDUCATION

**University of California, Davis**, B.S. Computer Science

**Expected Jun 2028**

**Coursework:** Data Structures & Algorithms (C++), Object-Oriented Programming (C++), Assembly Programming

## EXPERIENCE

**AI Intern**, Demolish Foods

**Aug 2026 – Sep 2026**

- Built an evidence-linked R&D knowledge system end to end (FastAPI, PostgreSQL, React/TypeScript) that lets food scientists search past experiments, compare measurements, and trace every answer back to its source document
- Designed an ingestion pipeline turning a large Drive archive into structured, human-reviewed experiment records
- Combined deterministic extraction with LLM-assisted Q&A that cites sources and flags missing or conflicting data
- Secured the system with Google SSO (OIDC), role-based access tiers, and a read-only Drive connector; backed it with 1,900+ automated tests (pytest, Playwright) and presented a live demo to the R&D team

**Undergraduate Research Assistant**, Z-Lab, UC Davis

**May 2026 – Present**

- Built a sleep-staging pipeline for whole-night intracranial EEG (14 patients, 32K+ scored epochs, 100+ GB)
- Developed a gradient-boosting sleep-stage classifier that reached Cohen's  $\kappa$  0.56 vs. 0.39 for the published SleepSEEG algorithm (3-class Wake/NREM/REM, leave-one-subject-out CV)
- Traced anomalous benchmarks to a feature-definition error and a ground-truth label mismatch, fixing the evaluation

**Founder**, Efferent Systems

**Jun 2025 – Present**

- Lead a 5-person team building PyBCI, an open-source desktop platform (Python, PyQt5, FastAPI) for EEG acquisition, preprocessing, visualization, and ML model training; released v1.0, now used by UC Davis Neurotech club members
- Set up GitHub Actions CI/CD (tests, release builds, SDK publishing) and a ~940-test pytest suite for the desktop app
- Built SandBox, a headless harness that replays PyBCI pipelines on synthetic EEG to catch regressions hardware-free

## PROJECTS

**Second Brain:** Always-On Personal AI Agent

Python, Claude Code, SQLite, discord.py

- Architected an agent over a 1,100+ note Obsidian knowledge graph that ingests Gmail, Discord, WhatsApp, Google Calendar/Drive, and browser activity into a SQLite FTS5 store, triaging it into tasks, events, and opportunities
- Built a Discord bridge (discord.py, systemd) to query it from any device: slash commands run headless Claude Code with read-only tools and return answers citing source notes, locked to a single-user allowlist
- Runs nightly on an always-on server: comms scan → contact graph → inbox triage → daily brief, with self-review

**AutoBuild:** Multi-Agent Software Engineering Orchestrator

Python, FastAPI, SQLite, Podman

- Built an orchestrator that runs feature requests through PM, tech-lead, engineer, and QA agent roles (Claude Code, Codex, DeepSeek) with human approval gates and tamper-evident audit records
- Sandboxed generated code in network-isolated rootless Podman and added deterministic LLM-output replay for CI; 1,300+ tests across ~45K lines of Python

**Meridian:** Semantic Graph Platform

TypeScript, React, SQLite

- Built a platform that parses any source (Markdown notes, code repositories) into one universal semantic graph, then diffs, validates, and visualizes it in a React studio; 15-package monorepo with a versioned plugin API and 2,400+ tests

**Kubera:** Algorithmic Trading Research System

Python, Claude Agent SDK, pandas

- Built a paper-trading arena that ranks strategy variants on live market data by the lower bound of a bootstrap 95% CI on expectancy, with walk-forward validation, backtest-gated Claude Agent SDK agents, and 600+ tests

## LEADERSHIP

**Board Member & Projects Division Lead**, UC Davis Neurotech

**2025 – Present**

- Lead the projects division; run technical design reviews of member BCI projects (paradigms, features, overfitting)
- Leading NeuralVLA, a supernumerary (extra) robotic arm controlled by EEG motor intent and vision-language-action (VLA) models; built a MuJoCo 7-DOF simulation and demonstration dataset
- Assembled and led a 5-person team at Global NeuroHack 2026, building an EEG + fNIRS speech decoder with contrastive cross-modal embeddings and an abstaining prototype classifier (PyTorch)

## SKILLS

**Languages:** Python, C++, TypeScript, SQL

**ML / Data:** TensorFlow, scikit-learn, NumPy, pandas, SciPy, MNE (EEG), LLM agents (Claude Agent SDK), RAG

**Backend / Frontend:** FastAPI, PostgreSQL, SQLite, React, Vite, PyQt5

**Tools:** Git, GitHub Actions, Docker, Podman, Linux/systemd, Google Cloud, MuJoCo, pytest, Playwright